

# UMAMAHESWAR RAJU SAGIRAJU

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## Professional Summary

Software Engineer with experience of designing and building scalable backend systems, RESTful APIs, and cloud-native microservices in Java and Python. Strong foundation in system design, database optimization, containerized deployments, and CI/CD automation. Hands-on experience building Agentic AI and RAG-based applications using LangChain and LangGraph, bridging traditional backend engineering with modern AI system development. Comfortable working across the full SDLC, from architecture and debugging to production deployment on Docker and Kubernetes.

## Technical Skills

- **Languages:** Java, Python, SQL, JavaScript
- **Backend & APIs:** Spring Boot, Spring MVC, FastAPI, RESTful APIs, Microservices Architecture, React, Node.js
- **Cloud & DevOps:** Docker, Kubernetes, Git, GitHub
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB
- **AI & Agentic Systems:** LangChain, LangGraph, RAG Pipelines, Prompt Engineering
- **RPA & Automation:** UiPath (*Associate Certified*), Blue Prism, Automa (*IPA Architect*)

## Professional Experience

### Software Engineer

*Tecklytics*

**Jul 2024 – Present**

*Bangalore, Karnataka*

*Backend Development & Data Extraction (Java + Spring Boot) — Client: Singapore*

- Designed scalable Java/Spring Boot microservices and RESTful APIs, improving system availability and reducing integration downtime for internal and external enterprise systems.
- Optimized Spring Data JPA/Hibernate database interactions to improve query performance and reduce latency across high-traffic services.
- Strengthened Jenkins/Maven CI/CD pipelines, streamlining build, test, and deployment automation and reducing production release risk.
- Resolved recurring production defects through hands-on debugging and troubleshooting, collaborating closely with QA and frontend engineers.

*System Modernization & Version Migration — Singapore Health Services*

- Reduced bot-initialization time across a large fleet of virtual machines by refactoring legacy Blue Prism VBOs (Visual Business Objects) and resolving technical debt during a large-scale RPA platform migration.
- Delivered a zero-downtime cutover for critical production automation workflows by orchestrating a parallel migration strategy with redundant environments and database clones.
- Implemented an upgraded authentication and security gateway, improving system observability and operational reliability.

*AI-Powered Code Analysis & RPA Review System — React, FastAPI, Google Gemini, PostgreSQL*

- Architected an automated analysis engine to evaluate complex RPA workflows using LLM-based pattern recognition, reducing manual code review effort.
- Achieved 100% data-processing integrity by engineering fault-tolerant retry logic and recovery algorithms against malformed automation service outputs.
- Accelerated technical troubleshooting and decision-making via a reactive visualization layer for nested activity trees and complexity metrics across large RPA workflow sets.

## Projects

### Simple Social – Multimedia Storage Platform

[multimedia-storage-sharing-app](#)

*Tech Stack: Python, FastAPI, Streamlit (Frontend), SQLAlchemy, aioredis, FastAPI-Users, JWT, ImageKit.io (Cloud Storage)*

- Built a full-stack social media platform for sharing images and videos using an asynchronous FastAPI backend and Streamlit-based interactive frontend.
- Implemented JWT-based authentication and session management with FastAPI-Users, covering registration, login, and protected route access.
- Integrated ImageKit for cloud-native media storage and on-the-fly transformation, including dynamic text overlays and blurred background padding for uploaded images and videos.
- Designed async database models with SQLAlchemy and aioredis to manage users and posts, and enforced owner-based access control on post decisions.

### Job Application Management System – Backend

*Java 11, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, REST APIs, Maven*

- Engineered an enterprise-grade backend system for job and company management following industry-standard RESTful microservices design principles.
- Modeled complex relational databases and optimized data-layer mapping utilizing JPA entity relationships and Hibernate.

## Education

### B.Tech, Information Science and Engineering

*Sai Vidya Institute of Technology, Bangalore, Karnataka*

**2019 – 2023**